

Module 3: Modern Engineering Materials

Crystal structure, Structure of materials, Metallic glasses, Liquid Crystals, Shape memory alloy and Biomaterials, Properties of materials, Transforming materials, Structure and transformation of materials, Composite materials and smart materials, Engineering applications of materials.

BIOMATERIALS

Any material, natural or man-made, that comprises whole or part of a living structure or Biomedical device that performs, augments, or replaces a natural function.

- Biomaterials are used to make devices that would be in close or direct contact with the body to augment or replace faulty materials.
- Biomaterials have had a major impact on the contemporary medicine and patient care.
- Used: artificial skin, artificial blood vessels, total artificial hearts, pacemakers, dental fillings, wires, plates and pins for bone repair, total artificial joint replacements.

- Biomaterials can be metals, ceramics, polymers, glasses, carbons, and composite materials.
- The field of biomaterials uses ideas from medicine, biology, chemistry, materials science and engineering.
- Engineers play a major role in this field, but they have to work closely with synthetic chemists to optimize materials properties and physicians to ensure that the device is useful in clinical applications.

Finger joint



Breast implant



Heart valve



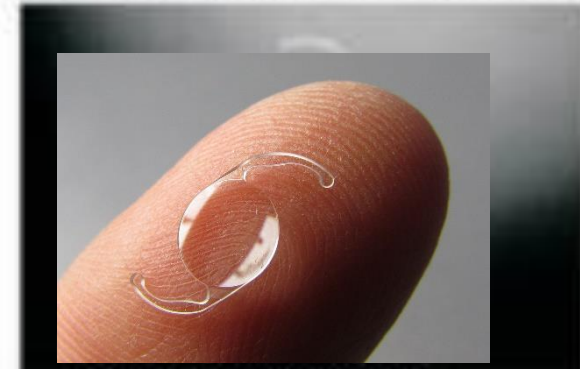
Hip joint



Artificial heart



Intraocular lens (IOL)



Criteria:

What makes a biomaterial, biomaterial

Biocompatibility:

Biocompatibility is the ability of a material to perform with an appropriate host response in a specific application.

“Appropriate host responses” include lack of blood clotting, resistance to bacterial colonization and normal healing.

Toxicity:

A biomaterial should not be toxic, unless it is specifically engineered for such requirements.

Mechanical and Performance Requirements:

Biomaterials and devices have mechanical and performance requirements that originate from the physical (bulk) properties of the material. There are three categories of such requirements: mechanical performance, mechanical durability and physical properties.

Types of Biomaterials

Metallic Biomaterials:

Metal : **load-bearing implants**, such as hip and knee prostheses and fracture fixation wires, pins, screws and plates.

3 Subgroups of Metallic Biomaterials

- stainless steels,
- the cobalt-based alloys,
- titanium metals and alloys



- These materials have biocompatibility, appropriate mechanical properties, corrosion resistance and reasonable cost.
- Useful for binding the fractured bone, **do not corrode** and **do not release harmful toxins** when exposed to **body fluids** and therefore can be left inside the body for a long period of time.

Ceramic and Glass Biomaterials

- Used as components of hip implants, dental implants, middle ear implants, and heart valves.
- biomedical applications: alumina, zirconia, bioglass, tricalcium phosphate.
- Although they do not undergo corrosion, ceramics and glasses are susceptible to other forms of degradation when exposed to the physiological environment.

Drawbacks

- The major drawbacks to the use of ceramics and glasses as implants are their **brittleness** and **poor tensile properties**.
- Ceramics and glasses **fail** at low stress when loaded in tension or bending.

Polymeric Biomaterials

- Used for making cardiovascular devices and for replacement and augmentation of various soft tissues.
- vascular grafts, heart valves, artificial hearts, breast implants, contact lenses etc.
- Biodegradable polymers are used for sutures, controlled drug delivery, tissue engineering, and fracture fixation.

Drawbacks

- Compared with metals and ceramics, polymers have much lower strengths and moduli but they can be deformed to a greater extent before failure.
- Polymers are generally not used in biomedical applications that bear loads (such as body weight).

Titanium and Titanium Alloys

- Titanium and some of its alloys are used as biomaterials for dental and orthopaedic application.
- These materials are biologically inert biomaterials.
- excellent corrosion resistance
- they do not produce any adverse reactions and are tolerated well by the human body.
- Titanium and its alloys possess suitable mechanical properties such as strength, bend strength and fatigue resistance.
- Ti-6Al-4V alloy has been used for decades as an implant material due to its excellent properties.
- Vanadium is a potentially toxic element and as an alternative are the Ti-6Al-7Nb alpha-beta alloy, Protasul-100